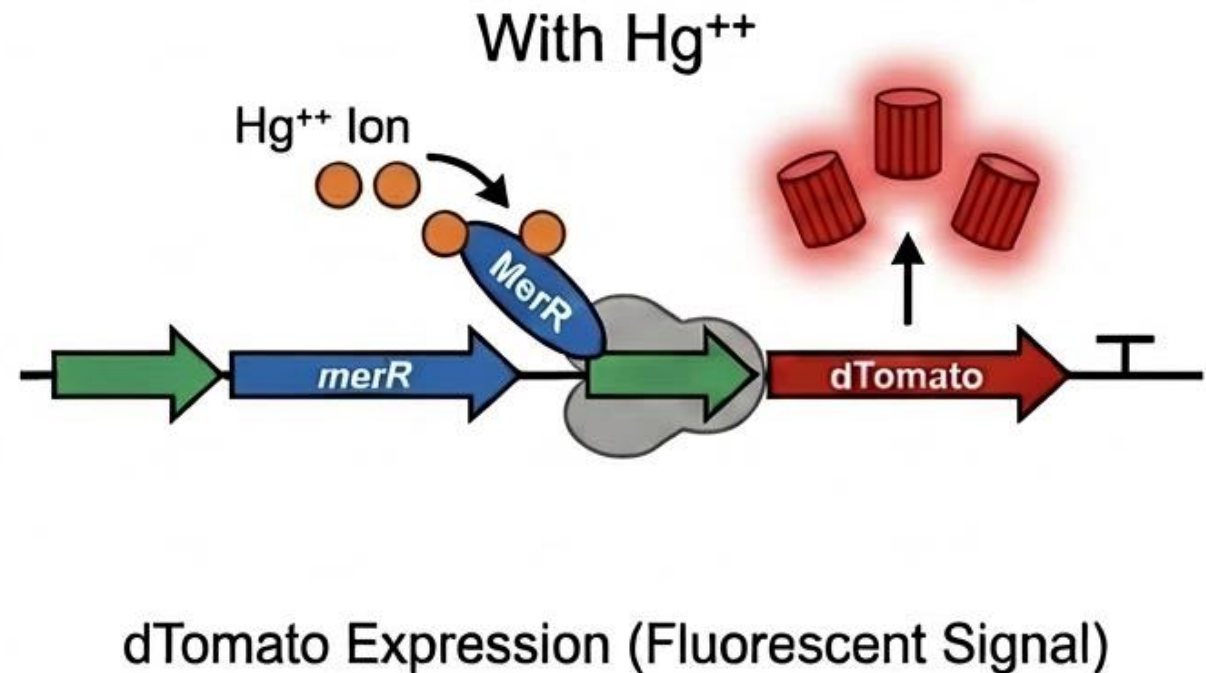
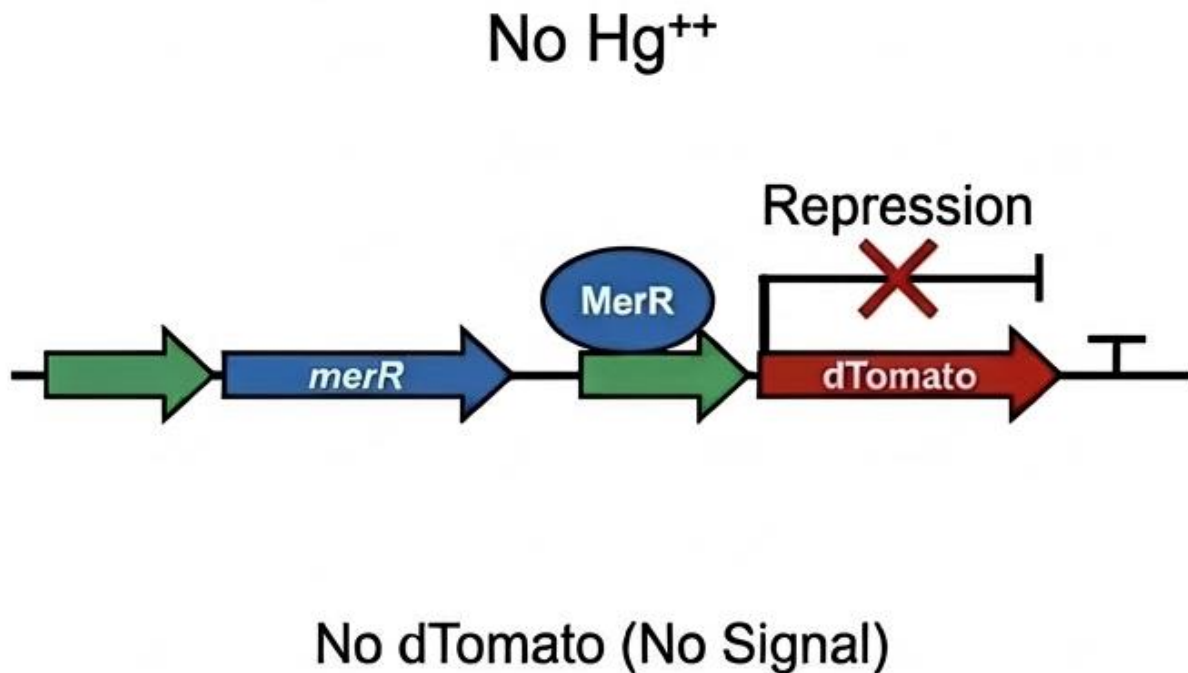
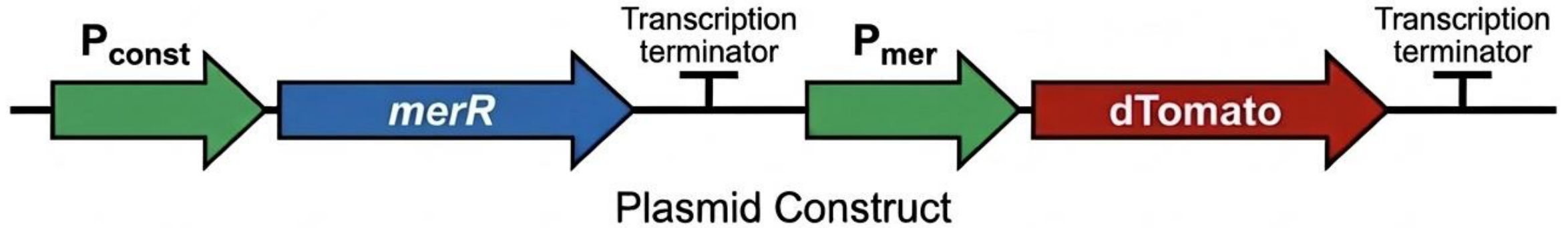


Protein-based transcriptional regulator (MerR/pMer)



Protein-based transcriptional regulator (MerR/pMer)

Two transcription units:

- **Pconst → merR**: a constitutive promoter that always expresses the MerR regulator protein, followed by a terminator.
- **Pmer → dTomato**: the mercury-responsive promoter driving the red fluorescent reporter, followed by a terminator.

No Hg^{2+} (OFF)

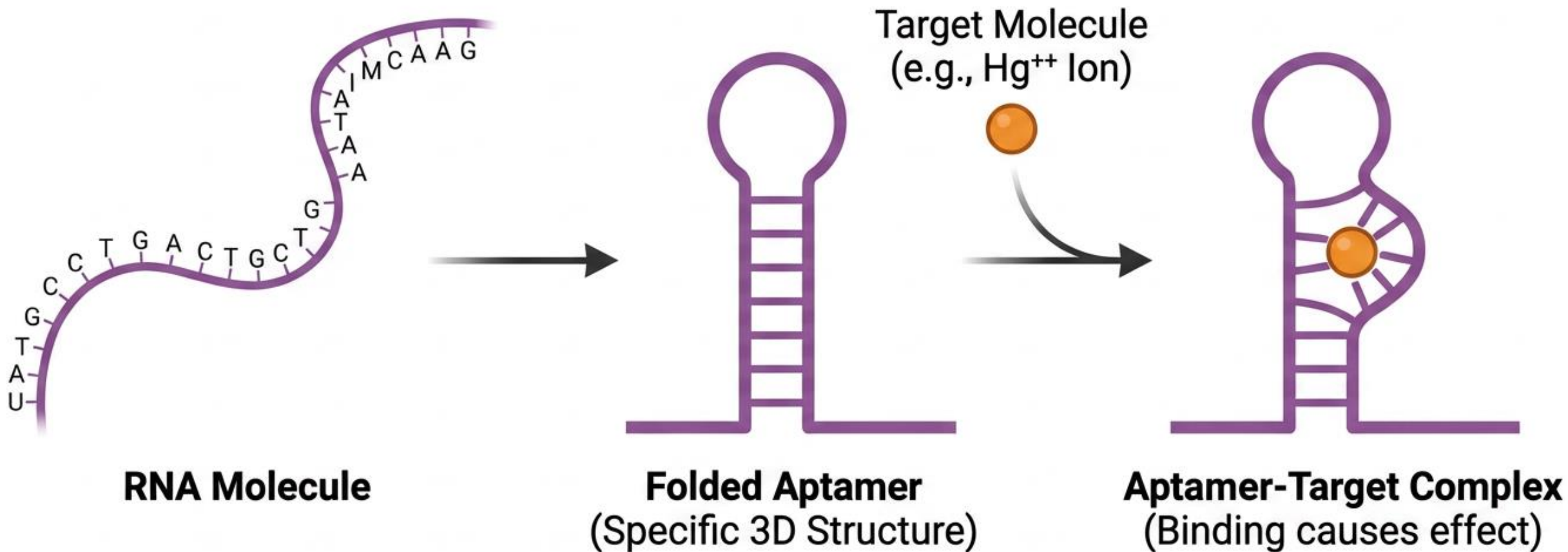
MerR protein binds the Pmer promoter and represses it, blocking transcription. No dTomato is made → no fluorescent signal.

With Hg^{2+} (ON)

Hg^{2+} ions bind MerR, causing a conformational change. Instead of blocking, MerR now recruits/activates RNA polymerase (grey shape) at Pmer, turning on transcription.

dTomato is produced → bright red fluorescence signals mercury contamination.

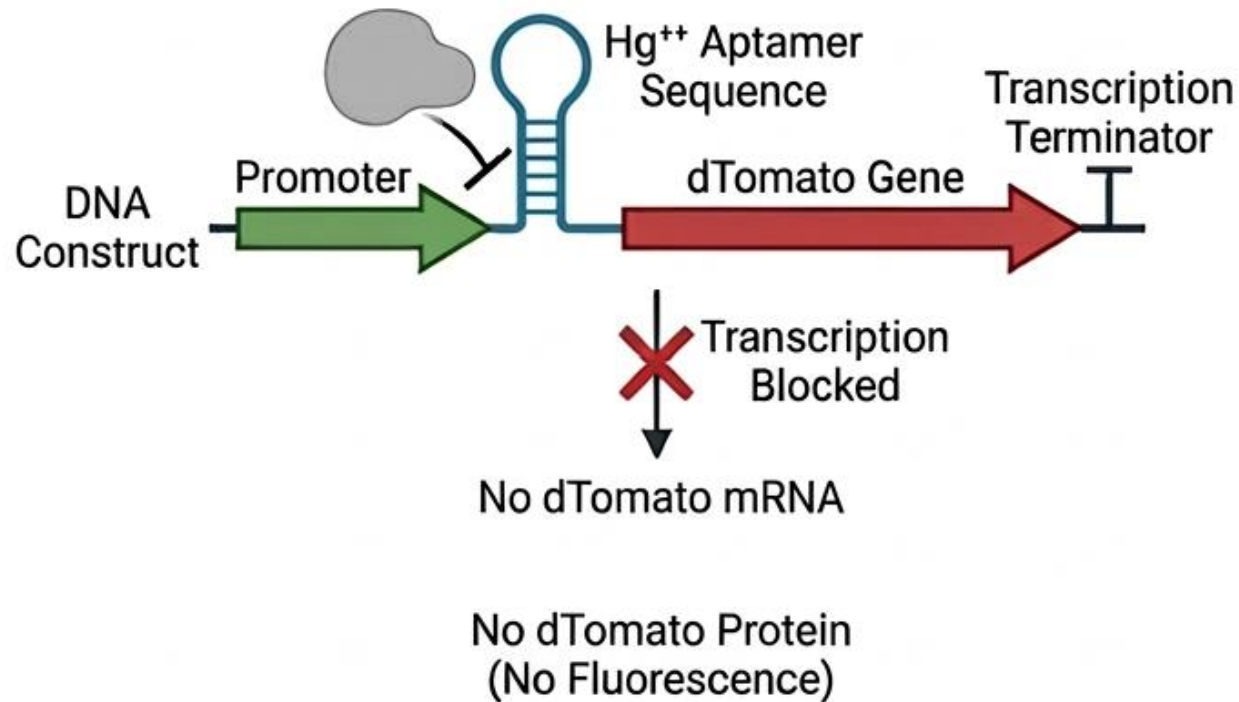
What is an Aptamer?



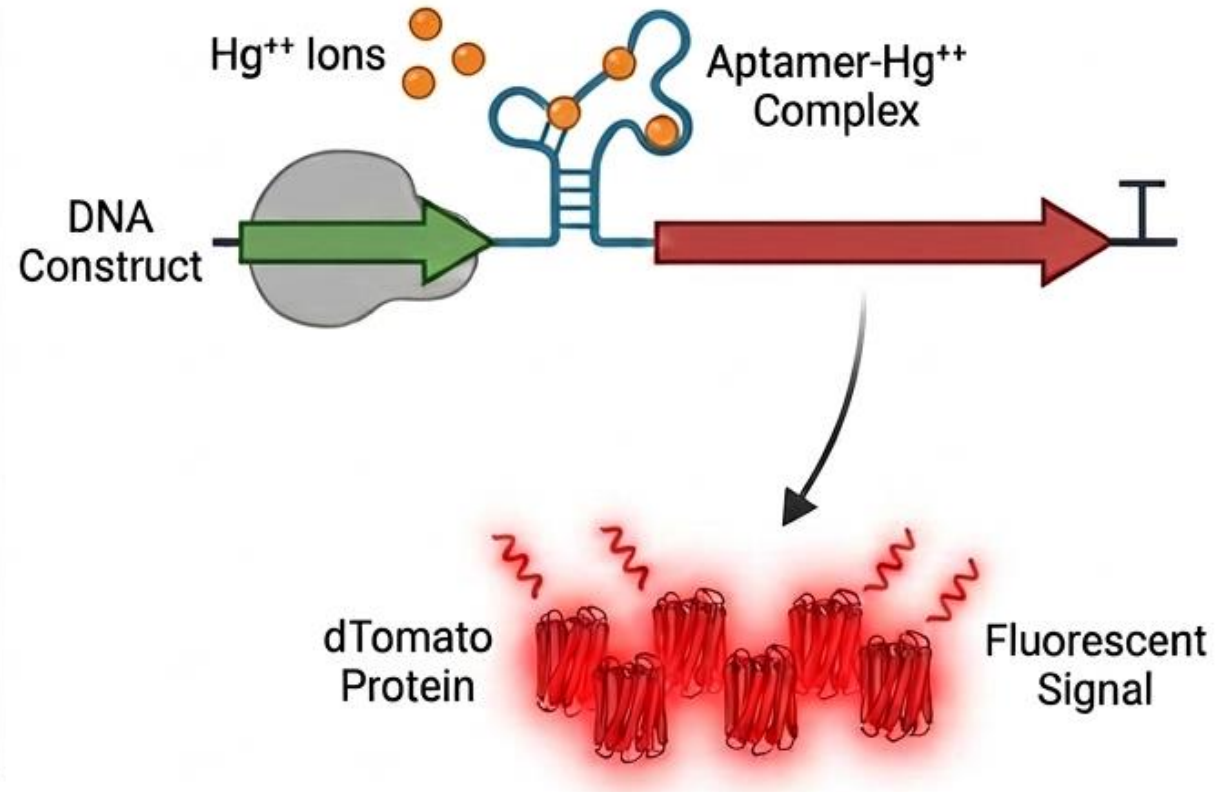
Aptamers are short single-stranded DNA or RNA molecules that can bind to specific target molecules with high affinity and specificity, often undergoing a conformational change upon binding to exert a regulatory function.

Aptamer-based Mercury ion (Hg^{++}) Detection

No Hg^{++} (No Signal)



With Hg^{++} (Fluorescent Signal)



Aptamer-based Mercury ion (Hg^{2+}) Detection

Construct design:

- Promoter $\rightarrow$ Hg^{2+} aptamer sequence $\rightarrow$ dTomato gene $\rightarrow$ transcription terminator.
- The aptamer forms a hairpin/stem-loop structure right after the promoter.

No Hg^{2+} (OFF)

Without mercury, the aptamer folds into a stem-loop that blocks transcription.
No dTomato mRNA $\rightarrow$ no dTomato protein $\rightarrow$ no fluorescence

With Hg^{2+} (ON)

Hg^{2+} ions bind the aptamer, forming an aptamer- Hg^{2+} complex that changes its shape. This rearrangement clears the block, so RNA polymerase transcribes the gene.

dTomato mRNA is produced $\rightarrow$ dTomato protein is made $\rightarrow$ red fluorescent signal